

Bitwise linear trail through **ARADI** found by SAT

CiVerLy

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1 ARADI

Maximal linear correlation: 2^{-25}

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	0	2	1	0	0	0	4	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	0	2	1	0	0	0	4	0	
ARADI ← round0																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	0	4	1	0	0	2	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	0	4	1	0	0	2	0	0	
ARADI ← round1																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	
ARADI ← round2																																		
0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	
0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	
ARADI ← round3																																		
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	0	8	0	8	0	0	1	0		
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	0	8	0	8	0	0	1	0		
ARADI ← round4																																		
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0			
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0			
FunctK																																		
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0			
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0			

2 ARADI-round0

Maximal linear correlation: $2^{-9.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	2	1	0	0	0	4	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	2	1	0	0	0	4	0
<i>BN</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	2	1	0	0	0	4	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	2	1	0	0	0	4	0
<i>SBiasLayer</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	8	6	1	2	5	2	4	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	8	6	1	2	5	2	4	0
<i>LinearLayer0</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0

3 SBoxLayer

Maximal linear correlation: $2^{-9.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	0	2	1	0	0	0	4	0
																															
2	0	0	0	0	2	1	0	0	0	0	1	0	0	2	0	0	2	0	2	0	0	2	0	0	1	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
3	0	0	0	0	3	1	0	0	0	0	1	0	0	3	0	0	3	0	3	0	0	3	0	0	1	0	0	0	0	0	0
																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	8	6	1	2	5	2	4	0

4 LinearLayer0

Maximal linear correlation: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	8	6	1	2	5	2	4	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	4	0	2	5	2	0	0	8	6	1	2	5	2	4	0
L0								L0								L0								L0							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0

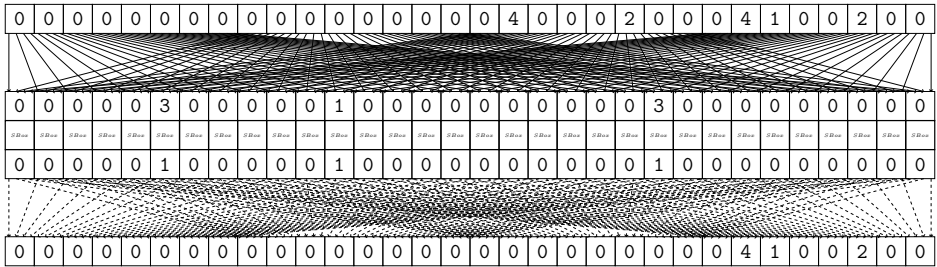
5 ARADI-round1

Maximal linear correlation: $2^{-3.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
<i>BN</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	2	0	0	0	4	1	0	0	2	0	0
<i>SBiasLayer</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	1	0	0	2	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	1	0	0	2	0	0
<i>LinearLayer1</i>																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

6 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



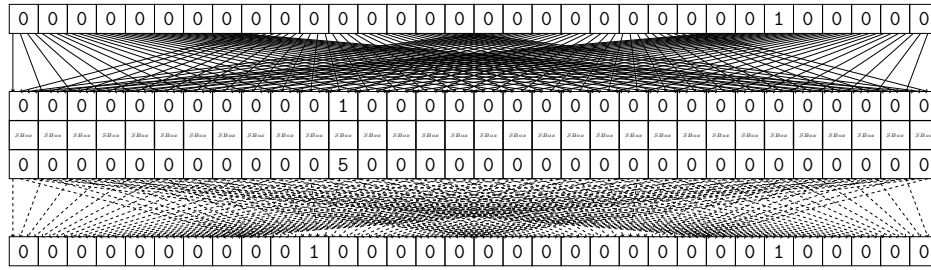
7 LinearLayer1

Maximal linear correlation: $2^{-0.0}$ [illegible]

8 ARADI-round2

Maximal linear correlation: $2^{-1.0}$ [illegible]

9 SBoxLayer

Maximal linear correlation: $2^{-1.0}$ 

10 LinearLayer2

Maximal linear correlation: $2^{-0.0}$ [illegible]

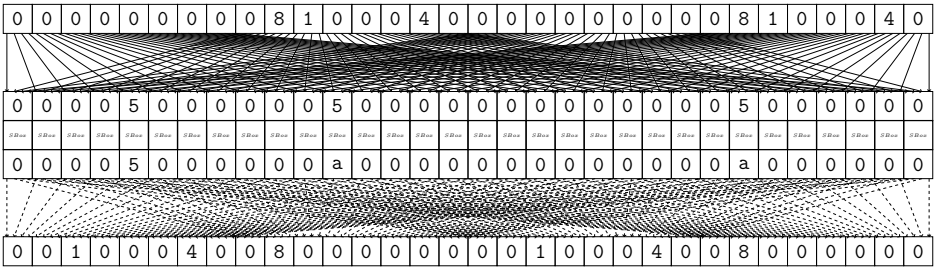
11 ARADI-round3

Maximal linear correlation: $2^{-3.0}$

0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0
0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0
RK																																
0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0
0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	4	0
SBoxLayer																																
0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0
0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0
LinearLayer3																																
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	0
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	0

12 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



13 LinearLayer3

Maximal linear correlation: $2^{-0.0}$

0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	
0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	0	0	0	1	0	0	0	4	0	0	8	0	0	0	0	0	
L3								L3								L3								L3							
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0

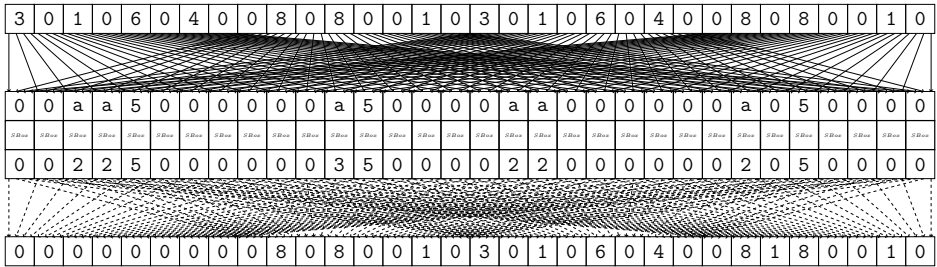
14 ARADI-round0

Maximal linear correlation: $2^{-9.0}$

3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0
<i>SK</i>																															
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0
3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	0	8	0	0	1	0
<i>SBoxLayer</i>																															
0	0	0	0	0	0	0	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	1	8	0	0	1	0
0	0	0	0	0	0	0	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	1	8	0	0	1	0
<i>LinearLayer0</i>																															
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0

15 SBoxLayer

Maximal linear correlation: $2^{-9.0}$



16 LinearLayer0

Maximal linear correlation: $2^{-0.0}$

0	0	0	0	0	0	0	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	1	8	0	0	1	0
0	0	0	0	0	0	0	0	0	8	0	8	0	0	1	0	3	0	1	0	6	0	4	0	0	8	1	8	0	0	1	0
L0								L0								L0								L0							
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0
0	0	0	0	0	0	0	0	1	9	0	9	2	2	3	0	7	2	7	6	a	8	0	c	1	b	1	9	2	2	7	0